

Tutorial

Modern Physics

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For exercises from the textbook, the corresponding number is reported. Text may have been edited, for clarity.
Unless otherwise specified, all results are given with three significant digits, avoiding rounding errors.

The solutions are provided on an "as is" basis, without warranty. Any error report is appreciated.

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WEEK I

CHAPTER 3. THE EXPERIMENTAL BASIS OF QUANTUM PHYSICS

• **Thompson experiment / Example 3.1**

In an experiment similar to Thomson's, we use deflecting plates $\ell = 5.0$ cm in length with an electric field $E = 1.2 \times 10^4$ V/m. Without magnetic field, we find an angular deflection of $\theta = 30^\circ$, and with a magnetic field $B = 8.8 \times 10^{-4}$ T we find no deflection. What is the initial velocity of the electron and its charge to mass ratio q/m ?

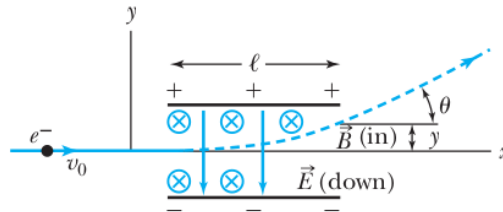


Figure 1: Thomson's experiment: the ratio e/m can be determined from E , B , θ , and ℓ .

• **Energy of a light pulse / Ex. 3.38**

The human eye is sensitive to a pulse of light containing as few as $N_\gamma = 100$ photons. For the orange light of wavelength $\lambda = 610$ nm, how much energy is contained in the pulse?

• **Photoelectric effect / Ex. 3.35**

What is the maximum wavelength $\lambda_{\max}$ of incident light that can produce photoelectrons from silver, which has work function $\phi = 4.64$ eV? What will be the maximum kinetic energy of the photoelectrons if the wavelength is halved, i.e., $\lambda' = \lambda_{\max}/2$?

• **Diffraction/ Ex. 3.10**

Light from a slit passes through a transmission diffraction grating of 400 lines/mm, which is located at $\ell = 3.0$ m from a screen. What are the distances on the screen (from the unscattered slit image) of the three brightest visible (first-order) hydrogen lines?

• **Blackbody radiation / Ex. 3.17**

Calculate $\lambda_{\max}$ for blackbody radiation for (a) liquid helium ($T = 4.2$ K), (b) room temperature ($T = 293$ K), (c) a steel furnace ($T = 2500$ K), and (d) a blue star ($T = 9000$ K).

• **Human radiation / Ex. 3.20**

(a) At what wavelength will the human body radiate the maximum radiation? (b) Estimate the total power radiated by a person of medium build (assume an area given by a cylinder of height $h = 175$ cm and radius $r = 13$ cm). (c) Using your answer to (b), compare the energy radiated by a person in one day with the energy intake of a $\mathcal{E} = 2000$ kcal diet.

• **Line spectra of H emitted by a moving star / Ex. 3.14**

Suppose that a detector in a new space telescope will be capable of detecting visible light in the wavelength range of $\lambda_{\min} = 400$ nm to $\lambda_{\max} = 700$ nm. (a) List all the wavelengths for the hydrogen atom that are in this range and their series name. (b) The detector measures visible wavelengths of $\lambda_1 = 537.5$ nm, $\lambda_2 = 480.1$ nm, and $\lambda_3 = 453.4$ nm, that researchers believe are due to the hydrogen atom. Why are all the known visible hydrogen lines not detected? (c) Use these data to calculate the speed of the stellar object that emitted the spectra. Assume that the object is not rotating. Why might rotation be an issue?

4. STRUCTURE OF THE ATOM

• **Rutherford scattering / Ex. 4.11**

The nuclear radii of aluminum and gold are approximately $r_{\text{Al}} = 3.6 \text{ fm}$ and $r_{\text{Au}} = 7.0 \text{ fm}$, respectively. The radii of protons and alpha particles are $r_{\text{p}} = 1.3 \text{ fm}$ and $r_{\alpha} = 2.6 \text{ fm}$, respectively.

(a) What energy α particles would be needed in head-on collisions for the nuclear surfaces to just touch? —This is about where the nuclear force becomes effective, it can be neglected at larger distances, and consider only the Coulomb interaction.

(b) What energy protons would be needed?

For both (a) and (b), perform the calculation for aluminum and for gold.

5. WAVE PROPERTIES OF MATTER AND QUANTUM MECHANICS I

• **Bragg scattering / Ex. 5.19**

In an X-ray scattering experiment, an intense reflected beam is found at $\phi = 53^\circ$ for a crystal with an interatomic distance of $D = 0.23 \text{ nm}$, see the sketch (assuming for the sake of simplicity that $D_x = D_y = D$). What is the lattice spacing of the planes responsible for the scattering? Assuming first-order diffraction, what is the wavelength λ , frequency f , and energy E of the incident photons?

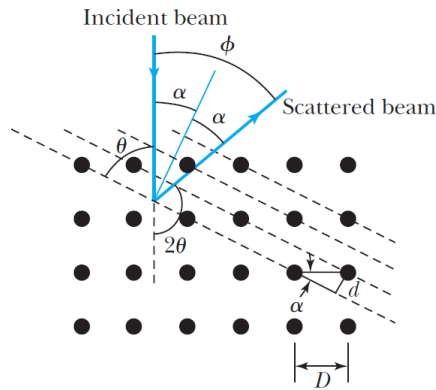


Figure 2: Bragg diffraction